Wi-Fi 8 Unveiled: Enhancing Spectrum Efficiency with Non-Primary Channel Access (NPCA)

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***Abstract*—The IEEE 802.11bn draft amendment to the 802.11 Wi-Fi standard introduces Non-Primary Channel Access (NPCA) to enhance spectrum efficiency and mitigate congestion in dense wireless environments. NPCA allows stations to transmit over non-primary channels when the primary channel is occupied, improving overall spectrum utilization and reducing transmission delays. This paper presents a detailed evaluation of NPCA, focusing on its effectiveness in scenarios involving hidden nodes—one of the key challenges limiting its performance. Our research models NPCA based on IEEE contributions and proposals, implementing realistic network topologies to analyze both its strengths and weaknesses. The findings provide critical insights into NPCA's practical deployment, highlighting its potential benefits and identifying areas requiring further optimization to ensure fairness and adaptability in real-world networks.**

I. Introduction

Wireless local area networks (WLANs) have evolved significantly to meet the growing demands for higher throughput, lower latency, and more reliable connectivity. At the core of this evolution is the Institute of Electrical and Electronics Engineers (IEEE) 802.11 family of standards, which has consistently introduced enhancements to improve network performance and service quality.

Previous amendments, including IEEE 802.11ax (Wi-Fi 6) [1] and IEEE 802.11be (Wi-Fi 7) [2], introduced key features to improve efficiency and throughput. Wi-Fi 6, also referred to as high efficiency (HE), focused on supporting a large number of concurrent stations (STAs) through orthogonal frequency division multiple access (OFDMA). It also introduced preamble puncturing and 8-by-8 downlink (DL) and uplink (UL) multi-user multiple-input multiple-output (MIMO) to support high-demand applications. Wi-Fi 7 further advanced efficiency with extremely high throughput (EHT) capabilities, supporting wider channels (up to 320 MHz) and multi link operation (MLO), enabling devices to utilize multiple frequency bands simultaneously. These enhancements enabled peak data rates exceeding 40 Gbps [3].

Despite the improvements in previous amendments, several challenges persist in modern wireless networks. Dense deployments with multiple STAs competing for limited spectrum result in frequent collisions and reduced throughput. Spectrum underutilization remains a common issue, particularly when primary channels are occupied by legacy STAs using narrower bandwidths. In high-density environments, overlapping basic service set (OBSS) interference further degrades channel access efficiency. Additionally, latency and packet loss are significant concerns, especially for delay-sensitive applications such as extended reality (XR).

To address these challenges, the latest amendment —IEEE 802.11bn, branded as ultra high reliability (UHR) [4], [5]— introduces advanced physical layer (PHY) and medium access control (MAC) features [6]. Innovations such as distributed resource unit (DRU), unequal modulation, seamless roaming, dynamic power save, multi-access point (AP) coordination, and non-primary channel access (NPCA) aim to enhance spectrum efficiency and mitigate interference, particularly in dense deployment scenarios. Its objectives include achieving a 25 % increase in data rates, a 25 % reduction in tail latency, and 25 % fewer lost MAC protocol data unitss (MPDUs).

Among the introduced features, NPCA stands out as a key enhancement aimed at improving spectrum utilization and reducing congestion. Traditionally, Wi-Fi devices could not transmit unless the primary 20 MHz channel was free, leading to underutilization of available spectrum. NPCA addresses this inefficiency by enabling both APs and STAs to access a non-primary channel when the primary channel is occupied by OBSS traffic. This allows a basic service set (BSS) to initiate transmission on a designated NPCA primary channel, ensuring better coexistence and overall network performance, particularly in environments where Wi-Fi 8 devices operate alongside legacy nodes using narrower bandwidths.

While NPCA enhances spectral efficiency and reduces latency, its implementation presents challenges, particularly in hidden node scenarios. If a device fails to detect a frame exchange initiating the NPCA switch, synchronization issues may arise [7]. Additionally, fairness concerns emerge in mixed deployments where some STAs lack NPCA capabilities, potentially disadvantaging legacy devices [8]. Addressing these challenges requires refined coordination mechanisms and adaptive policies to ensure equitable distribution of NPCA benefits across the network.

The contributions of this paper are as follows: Modeling NPCA based on IEEE contributions, proposals, and recent draft specifications, aligning with ongoing standardization efforts; Designing scenarios to evaluate NPCA under diverse network topologies and traffic conditions; Performing a comprehensive analysis of NPCA, focusing on the hidden node problem as a key efficiency challenge; Assessing performance under both ideal (full visibility) and hidden node conditions; Providing practical insights into NPCA deployment, highlighting key benefits and limitations. To the best of our knowledge, only a few articles have addressed NPCA so far, and none of them focus on the hidden node problem, which makes this work the first to provide an in-depth analysis of this aspect.

II. System Model

Network Topology: The system model considers a Wi-Fi 8 deployment consisting of multiple BSSs, each one including an AP and its associated STAs. All nodes are static —their positions do not change over time. The set of APs is denoted as $\mathcal{A} = \{A_1, \ldots, A_a, \ldots, A_A\}$, where each A_a corresponds to the AP of BSS bss_a. Similarly, each AP manages a set of STAs, defined as $\mathcal{S}_a = \{S_{a,1}, \ldots, S_{a,s}, \ldots, S_{a,S}\}$, where $S_{a,s}$ represents the s-th STA associated with AP A_a. To generalize, the complete set of STAs in the network is given by $\mathcal{S} = \bigcup_{a \in \mathcal{A}} \mathcal{S}_a$. The total set of network nodes, including both APs and STAs, is defined as $\mathcal{N} = \mathcal{A} \cup \mathcal{S} = \{N_1, N_2, \ldots, N_N\}$.

Spectrum Usage: The BSSs in the system are overlapping —multiple BSSs operate within the same frequency range, resulting in potential interference. While some BSSs utilize wider channels (e.g., 160 MHz), others operate on narrower bandwidths (e.g., 80 MHz). Despite these differences, all BSSs share overlapping primary channels, which leads to increased channel contention and coexistence challenges.

CSMA/CA and Signal Detection: Wi-Fi networks use the carrier sense multiple access/collision avoidance (CSMA/CA) mechanism to regulate channel access. A key component of CSMA/CA is the clear channel assessment (CCA), which determines whether a node perceives the medium as busy. If the medium is busy, the node defers transmission to avoid collisions [1]. To determine channel occupancy, CCA relies on two detection mechanisms:

- *Energy detection (ED)*, which considers the medium occupied if the received signal power exceeds the threshold $P^{\mathrm{ED}} = -62$ dBm, i.e., if $P_{n,m} > P^{\mathrm{ED}}$, then node n senses node m.
- *Packet detection (PD)*, which determines whether a successfully decoded 802.11 preamble is strong enough to consider the medium occupied, based on a threshold of $P^{\mathrm{PD}} = -82$ dBm. If $P_{n,m} > P^{\mathrm{PD}}$, then node n detects node m. In this case, the node can extract the network allocation vector (NAV), reserving the channel for the duration of the ongoing transmission.

If neither condition is met, the channel is considered available for transmission, but the possibility remains that hidden nodes may be active on the medium at the same time.

Connectivity and Sensing/Detection: To drive CSMA/CA and determine connectivity conditions, we define a connectivity matrix C, where each element $C_{n,m}$ indicates whether node n can sense and/or detect node m, where sensing and detection are determined by the ED P^{ED} and PD P^{PD} thresholds, respectively. In more details, matrix C is defined for all APs and STAs in the system. For a generic deployment with A APs and S STAs per BSS, the connectivity matrix is structured as:

$$C = \begin{bmatrix} C_{A_a,A_a} & C_{A_a,S_{a,1}} & C_{A_a,A_b} & \cdots & C_{A_a,S_{b,S}} \\ C_{S_{a,1},A_a} & C_{S_{a,1},S_{a,1}} & C_{S_{a,1},A_b} & \cdots & C_{S_{a,1},S_{b,S}} \\ C_{A_b,A_a} & C_{A_b,S_{a,1}} & C_{A_b,A_b} & \cdots & C_{A_b,S_{b,S}} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ C_{S_{b,S},A_a} & C_{S_{b,S},S_{a,1}} & C_{S_{b,S},A_b} & \cdots & C_{S_{b,S},S_{b,S}} \end{bmatrix}, \tag{1}$$

where $C_{n,m} = 1$ if node n detects node m via ED or PD, and $C_{n,m} = 0$ otherwise.

This matrix serves as a fundamental input to the system, defining connectivity conditions, guiding interference assessment, and supporting the evaluation of the NPCA feature.

RTS/CTS Protection: CSMA/CA alone is insufficient to mitigate the hidden node problem, where some nodes cannot detect ongoing transmissions, leading to frequent collisions. This occurs when a transmitter and receiver can communicate, but another node —unaware of the ongoing transmission— attempts to send data, thereby causing interference. To address this, the request-to-send (RTS)/clear-to-send (CTS) mechanism was introduced [1]. RTS/CTS begins with a transmitting node sending a short RTS frame to its intended receiver. If the receiver successfully receives this request, it responds with a CTS frame, granting the sender permission to transmit. This handshake ensures exclusive channel access before data transmission. Crucially, neighboring nodes that overhear either the RTS or CTS frame recognize that a transmission is about to occur and update their NAV. This prevents hidden nodes —who may not sense the original transmission— from interfering, as they remain silent for the reserved duration.

In our scenarios, all nodes implement RTS/CTS to improve coordination and reduce interference. Importantly, RTS/CTS will be used as trigger for NPCA activation.

III. NPCA Algorithm

Traditional Wi-Fi operation mandates that devices defer transmission whenever the primary channel is busy, even if other portions of their supported bandwidth remain idle [1]. NPCA is a feature introduced in Wi-Fi 8 to enhance spectrum efficiency, by allowing an AP and its STAs to access a non-primary channel when the primary channel is occupied by OBSS traffic [6]. This enables a reduction in latency and an increase in throughput, while maintaining fair contention.

Fig. 2 illustrates the NPCA alogrithm. In the following, we introduce the NPCA mechanism, whose process can be divided into the following phases:

A. NPCA Primary Channel Selection

Each BSS supporting NPCA has to designate a single NPCA primary channel [6]. This NPCA primary channel serves as the contention point for transmissions, when the primary channel is occupied by OBSS traffic. The NPCA primary channel must be within the AP's BSS operating channel width, and cannot be a punctured 20 MHz subchannel.

If an AP A_a as well as its associated STAs $\mathcal{S}_a = \{S_{a,1}, S_{a,2}, \ldots, S_{a,S}\}$ are NPCA-capable, A_a announces its NPCA primary channel configuration, ensuring that BSS bss_a operates under the same contention rules. Additionally, A_a specifies a predefined minimum time threshold, which determines the required duration of an OBSS transmission for triggering the transition to the NPCA primary channel. This threshold allows all devices within the bss_a to make consistent channel access decisions, ensuring fair and efficient spectrum utilization.

$$C = \begin{bmatrix} 1&1&1&1&1&1&1&1 \\ 1&1&1&1&1&1&1&1 \\ 1&1&1&1&1&1&1&1 \\ 1&1&1&1&1&1&1&1 \\ 1&1&1&1&1&1&1&1 \\ 1&1&1&1&1&1&1&1 \\ 1&1&1&1&1&1&1&1 \\ 1&1&1&1&1&1&1&1 \end{bmatrix}$$

(a) S1: Full Connectivity

$$C = \begin{bmatrix} 1&1&1&1&1&1&1&1 \\ 1&1&1&1&1&0&0&0 \\ 1&1&1&1&1&0&0&0 \\ 1&1&1&1&1&0&0&0 \\ 1&1&1&1&1&1&1&1 \\ 1&0&0&0&1&1&1&1 \\ 1&0&0&0&1&1&1&1 \\ 1&0&0&0&1&1&1&1 \end{bmatrix}$$

(b) S2: APs detect all, STAs only their AP

$$C = \begin{bmatrix} 1&1&1&1&1&1&0&0 \\ 1&1&1&1&1&1&0&0 \\ 1&1&1&1&0&0&0&0 \\ 1&1&1&1&0&0&0&0 \\ 1&1&0&0&1&1&1&1 \\ 1&1&0&0&1&1&1&1 \\ 0&0&0&0&1&1&1&1 \\ 0&0&0&0&1&1&1&1 \end{bmatrix}$$

(c) S3: Mixed connectivity

Fig. 1: Connectivity matrices for the three evaluated scenarios.

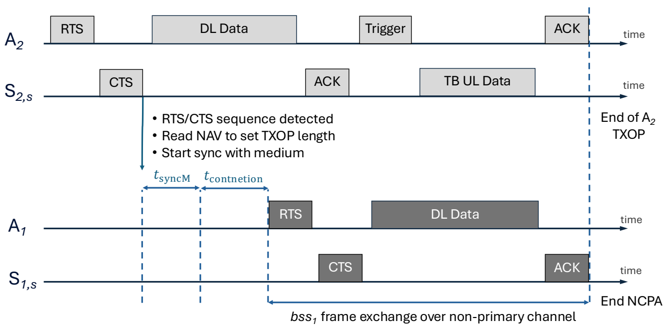


Fig. 2: NPCA algorithm diagram.

B. Triggering NPCA Operation

NPCA is activated when AP A_a or a STA $S_{a,n}$ detects an ongoing transmission on its primary channel, for example, through an OBSS RTS/CTS exchange. Upon detection, each NPCA-capable device evaluates the duration of the OBSS transmission, using the NAV from the control frames. If the estimated duration meets or exceeds the predefined threshold, announced during the NPCA primary channel selection phase, the device initiates a transition to the NPCA primary channel. This ensures that both the AP A_a and its STAs $\mathcal{S}_a$ can autonomously adapt to the spectrum conditions, maximizing channel utilization. Every transition to the NPCA primary channel requires devices to account for a switching delay, which is communicated between peers to maintain timing consistency [6]. The delay value is defined within a range of 0 to 256 μs, with a 4-μs resolution. To ensure synchronization, the AP A_a must adjust the initial RTS duration to accommodate the maximum switching delay among STAs from $\mathcal{S}_a$.

C. NPCA Transmission and Return to Primary Channel

Once the NPCA-capable AP A_a and its STAs $\mathcal{S}_a$ switch to the NPCA primary channel, they follow the standard enhanced distributed channel access (EDCA) contention rules [6]. To ensure fairness in access priority, NPCA operation uses the same EDCA parameters as the primary channel. In our evaluation, only the AP A_a contends for the NPCA primary channel while UL transmissions are governed by triggered UL transmissions, preventing uncoordinated UL access.

The transmission procedure follows these steps:

1) The device performs CCA on the NPCA primary channel to check if it is idle.
2) If the channel is idle, the device initiates an RTS/CTS handshake to reserve the medium.
3) Upon receiving a CTS response, the device transmits data within the allowed transmission opportunity (TXOP) duration, using an appropriate transmission scheme, such as single-user, multi-user MIMO, or multi-user OFDMA.
4) During transmission, the device ensures that the NPCA operation does not exceed the duration of the OBSS transmission on the primary channel.
5) When the NAV counter expires, the device ceases NPCA transmissions/receptions and switches back to the primary channel, resuming normal operation.

To maintain fairness and stability, the contention window (CW) and backoff counters for EDCA access are reset upon returning to the primary channel, ensuring consistent medium access behavior across both contention domains.

IV. NPCA Performance Evaluation

In this section, we evaluate the performance of NPCA based on the system model defined in Section II using customized, NS3-based system-level simulations. The evaluation considers three distinct scenarios, each involving two overlapping BSSs.

Each BSS consists of one AP and three associated STAs, denoted as $\mathcal{A} = \{A_1, A_2\}$ and $\mathcal{S}_a = \{S_{a,1}, S_{a,2}, S_{a,3}\}$, respectively. The nodes in bss_1 have a channel capability of 160 MHz, while the nodes in bss_2 operate within the lower 80 MHz of bss_1's channel. Thus, the primary channel of bss_1 (80 MHz) completely overlaps with the primary channel of bss_2, leading to contention and potential interference. Only bss_1 is NPCA-capable, meaning that all its APs and STAs can perform NPCA operations, whereas bss_2 follows traditional Wi-Fi contention rules. The NPCA primary channel is in the upper 80 MHz of bss_1. Simulations were conducted using fixed modulation and coding scheme (MCS), 9 and 11.

Scenarios: The key difference among the three tested scenarios lies in the visibility between nodes, which is determined by the connectivity matrix C (refer to Eq. 1). A value of 1 in C indicates that node n can sense or detect node m, while a value of 0 signifies that the nodes are hidden from each other. Fig. 1 illustrates the connectivity matrices for the three scenarios:

- `Scenario 1` This baseline scenario assumes full connectivity, where all nodes can detect the transmissions of all other nodes (see Fig. 1a).

- `Scenario 2` Here, APs can sense or detect signals from all nodes, whereas STAs can sense or detect all APs, but they remain hidden from the OBSS STAs (see Fig. 1b).
- `Scenario 3` In this scenario, APs detect their associated STAs, as well as the AP and one STA from the OBSS. Additionally, each BSS has one STA that is fully connected, detecting its AP, all STAs in its own BSS, and one STA and the AP from the OBSS. In contrast, the remaining STAs can only sense their AP and other STAs within the same BSS, but remain hidden from the OBSS (see Fig. 1c).

Evaluation Cases: For each scenario, three evaluation cases were examined to assess the impact of NPCA activation and the influence of control frame-end (CF-End) frames.

CF-End frames are control frames designed to announce that an AP has completed its data exchange, allowing the channel to be freed before the granted TXOP expires. When other devices detect this frame, they can immediately start contending for the channel.

The three evaluation cases are defined as follows:

- `Case 1` (No NPCA, No CF-End): Serves as benchmark, where neither NPCA nor CF-End frames are activated.
- `Case 2` (NPCA Activated): bss_1 operates with NPCA enabled, allowing assessment of its performance.
- `Case 3` (NPCA + CF-End Activated): NPCA is enabled in bss_1, while bss_2 employs CF-End frames to signal early channel release.

Offered Traffic: A first set of evaluations considers a full-buffer traffic model, where device queues are always filled with data to send. In this setup, each AP transmits six DL flows of 150 Mbps to its three associated STAs, while each STA generates an UL flow of 50 Mbps to its AP. All offered traffic belongs to the best effort (BE) access category.

Additionally, for `Scenario 1`, a low-load (under-saturation) simulation was conducted. In this case, each AP offered 20 Mbps per STA in the DL, while each STA transmitted at 7 Mbps in the UL. Further evaluations included an experiment where traffic increased linearly to analyze system behavior under different load conditions. This last experiment involved only DL traffic, with equal data rate per flow.

Evaluation Metrics: The primary key performance indicator (KPI) in this study is the mean throughput, measured as the aggregated DL, UL, and total (DL+UL) throughput per BSS, as well as the total network throughput. To further analyze NPCA operation, additional counters were evaluated to capture transmission and reception events on both the primary and non-primary channels, including the number of transmitted and received RTS frames, and the number of received MPDUs.

V. Results and Discussion

A. Scenario 1 Analysis

The results of the full-buffer simulation, when all nodes are fully connected, demonstrate the performance gain enabled

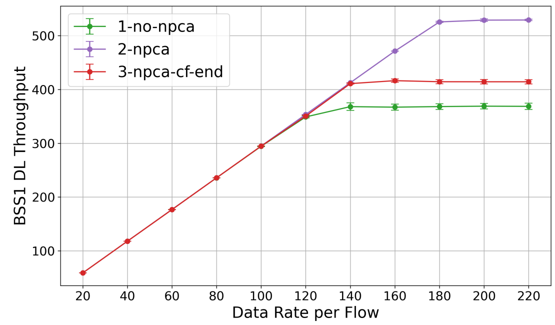


Fig. 3: `Scenario 1` bss_1 aggregated throughput for linearly increasing offered traffic, for MCS 9.

by NPCA. Fig. 4a compares the aggregated DL throughput across the three evaluation cases, while Fig. 4b presents the UL results. Tab. I summarizes the total aggregated throughput per BSS and for the entire system.

With NPCA enabled in bss_1 (without CF-End in bss_2), a 47 % DL throughput gain and a 20 % total throughput improvement are observed. When CF-End frames are used in bss_2, the performance gain from NPCA is reduced, but remains noticeable. This reduction occurs because CF-End frames, sent by A_2 on the primary channel, give bss_2 a contention advantage, as bss_1 is temporarily synchronized with the non-primary channel, and is thus easier for bss_2 to access the primary channel again for a new TXOP.

Additionally, Tab. III presents statistics on the amount of data frames received over the primary and non-primary channels for MCS 11, depending on the node and the evaluation case. Notably, enabling exchanges over the non-primary channel increases the total amount of delivered data. This gain is particularly evident for DL traffic, as STAs receive significantly more data frames when NPCA is activated. For instance, $S_{1,2}$ receives 56 % more MPDUs.

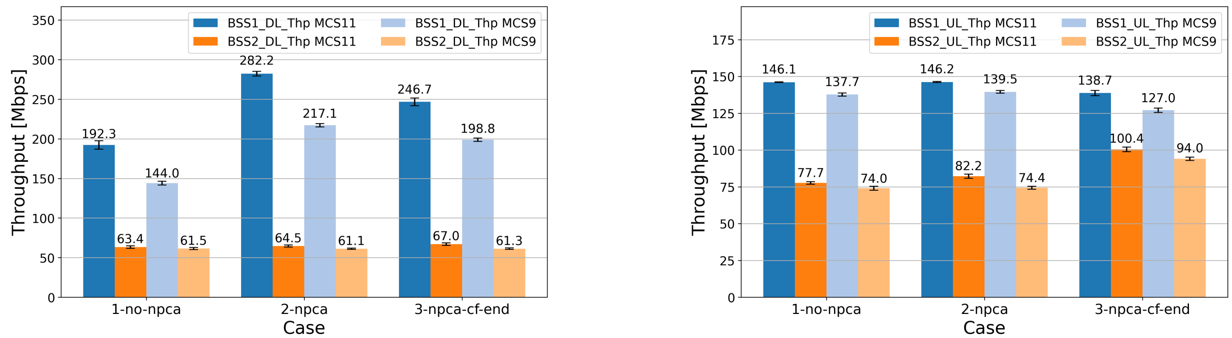
In the under-saturation scenario, NPCA enables successful data transmission over the non-primary channel, as shown in Tab. IV, (configuration with MCS 11). However, it does not result in any performance improvement. Since the available channel capacity is sufficient to accommodate all enqueued data, NPCA operation does not enhance throughput. Switching to the non-primary channel introduces overhead, including signaling and synchronization with a different part of the spectrum, which increases energy consumption. To improve

TABLE I: `Scenario 1` Aggregated throughput [Mbps].

	1. No NPCA		2. NPCA		3. NPCA + CF-End	
	MCS 11	MCS 9	MCS 11	MCS 9	MCS 11	MCS 9
bss_1	338.4	281.7	428.5	356.6	385.4	325.8
bss_2	141.1	135.6	146.7	135.5	167.4	155.3
Total	479.4	417.3	575.2	492.1	552.9	481.1

TABLE II: `Scenario 2` Aggregated throughput [Mbps].

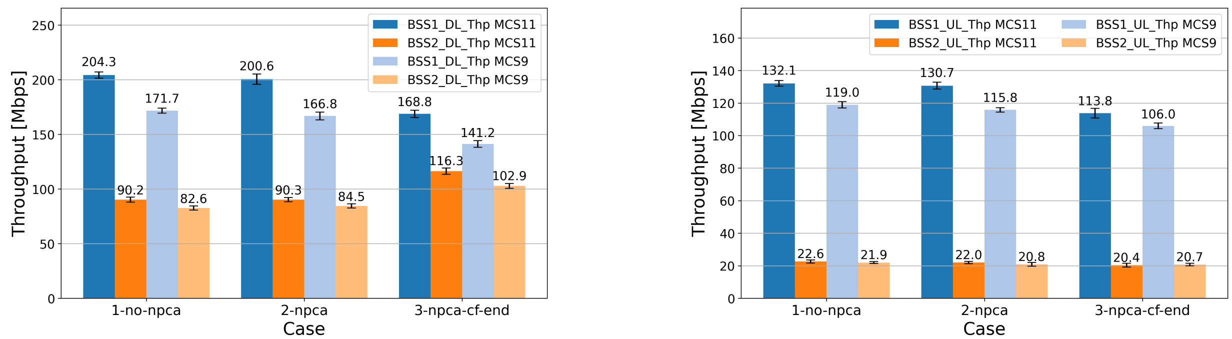
	1. No NPCA		2. NPCA		3. NPCA + CF-End	
	MCS 11	MCS 9	MCS 11	MCS 9	MCS 11	MCS 9
bss_1	336.4	290.6	331.2	282.6	282.5	247.2
bss_2	112.8	104.5	112.3	105.3	136.7	123.6
Total	449.2	395.2	443.5	387.9	419.2	370.8



(a) DL aggregated throughput.

(b) UL aggregated throughput.

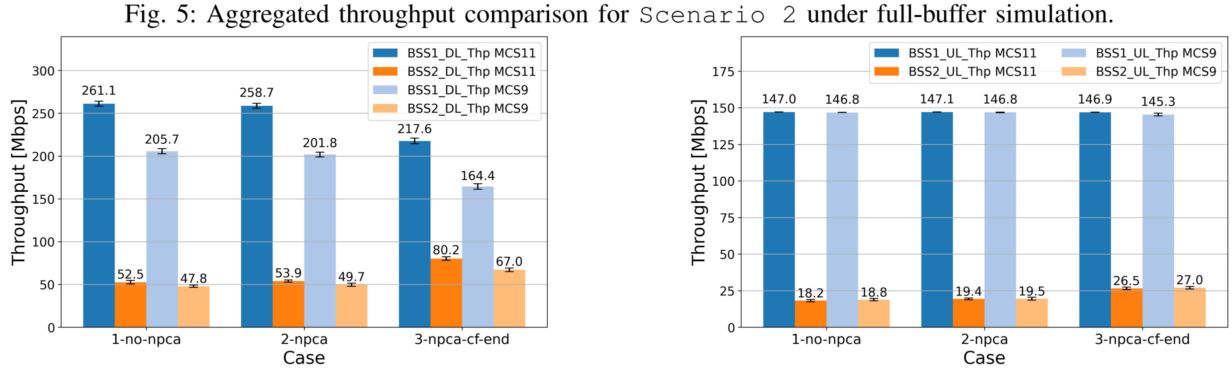
Fig. 4: Aggregated throughput comparison for `Scenario 1` under full-buffer simulation.



(a) DL aggregated throughput.

(b) UL aggregated throughput.

Fig. 5: Aggregated throughput comparison for `Scenario 2` under full-buffer simulation.



(a) DL aggregated throughput.

(b) UL aggregated throughput.

Fig. 6: Aggregated throughput comparison for `Scenario 3` under full-buffer simulation.

efficiency, future research should explore optimizing NPCA activation strategies to ensure it is triggered only when a noticeable performance gain can be achieved.

Fig. 3 also shows the performance comparison for three cases, when DL offered traffic increases linearly. It can be observed clearly that NPCA activation allows to reach higher aggregated throughput in bss_1, (where the feature is triggered). When, in addition to NPCA, the CF-End frames are activated in bss_2, the performance gain is reduced but still noticeable.

B. Scenario 2 Analysis

Since this setup includes hidden nodes, the results highlight a limitation of the NPCA feature. The DL aggregated throughput comparison is shown in Fig. 5a, while UL throughput is presented in Fig. 5b.

When NPCA is activated in bss_1, and bss_2 occupies the lower 80 MHz channel, A_1 switches to the upper 80 MHz channel. However, the STAs in bss_1 remain on the primary channel, as they cannot detect the full RTS/CTS sequence. Consequently, no successful exchange occurs over the non-primary channel, and NPCA activation neither improves nor degrades performance. Tab. II shows the detailed throughput.

Moreover, the use of CF-End frames in bss_2 has a mixed impact on network performance, slightly increasing bss_2 throughput while reducing the bss_1 one. As before, CF-End frames allow bss_2 to regain channel access more quickly, giving it an advantage. However, this creates a disadvantage for bss_1, which finds the channel occupied upon returning from the non-primary channel.

Additionally, an important observation is that A_1 continues sending RTS frames over the non-primary channel until its NAV (set by the OBSS) expires, This behavior leads to unnecessary energy consumption.

Tab. V presents additional NPCA-related KPIs for the configuration with MCS 11, showing that STAs in bss_1 do not receive any data over the non-primary channel, while A_1 continuously transmits RTSs over this part of the spectrum.

C. Scenario 3 Analysis

This scenario includes a mix of hidden and non-hidden nodes. When NPCA is activated in bss_1, and bss_2 utilizes the lower 80 MHz channel, A_1 can only detect the RTS/CTS sequence from bss_2 under specific conditions —when the exchange occurs between A_2 and $S_{2,1}$. Additionally, only $S_{1,1}$ in bss_1 is able to detect this exchange, and switch to the non-primary channel alongside A_1. As a result, overall utilization of the non-primary channel remains minimal.

Fig. 6a presents the DL aggregated throughput across all evaluated cases, while Fig. 6b shows the corresponding UL throughput. Although some successful transmissions occur between A_1 and $S_{1,1}$ over the non-primary channel, the overall performance gain due to NPCA remains negligible. Moreover, the use of CF-End frames in bss_2 negatively impacts total system throughput and reduces bss_1 performance, while marginally improving bss_2 throughput by enabling faster channel reoccupation.

Both `Scenario 2` and `Scenario 3` show that future research should focus on developing more advanced NPCA triggering mechanisms to mitigate resource wastage in the presence of hidden nodes.

VI. Conclusion

This article evaluated the NPCA feature in IEEE 802.11bn across various scenarios. When nodes are fully connected and detect RTS/CTS, NPCA improves DL throughput by up to 47 % and total throughput by 20 % under full-buffer traffic. In hidden node cases, these gains vanish due to failed control frame detection, leading to energy waste from unnecessary RTSs. Thus, while NPCA can boost spectrum efficiency, its impact depends on coordination and node detection. Future research should therefore focus on developing more advanced and adaptive NPCA triggering mechanisms. Future work should also include a quantitative analysis of energy consumption issue mentioned in the study.

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TABLE III: `Scenario 1` NPCA-related counters: received MPDUs in No NPCA vs. NPCA cases for full-buffer.

	1. No NPCA			2. NPCA			
	No. of Non-NPCA Rx MPDUs	No. of NPCA Rx MPDUs	**Total No. of Rx MPDUs**	No. of Non-NPCA Rx MPDUs	No. of NPCA Rx MPDUs	**Total No. of Rx MPDUs**	**The Gain of the No. of Rx MPDUs**
A_1	100228	0	**100228**	92401	10106	**102507**	**2%**
$S_{1,1}$	46450	0	**46450**	47435	23041	**70476**	**52%**
$S_{1,2}$	33271	0	**33271**	36668	15766	**52434**	**56%**
$S_{1,3}$	51132	0	**51132**	43717	25316	**69033**	**35%**

TABLE IV: `Scenario 1` NPCA-related counters: received MPDUs in No NPCA vs. NPCA cases, for under-saturation.

	1. No NPCA			2. NPCA			
	No. of Non-NPCA Rx MPDUs	No. of NPCA Rx MPDUs	**Total No. of Rx MPDUs**	No. of Non-NPCA Rx MPDUs	No. of NPCA Rx MPDUs	**Total No. of Rx MPDUs**	**The Gain of the No. of Rx MPDUs**
A_1	14234	0	**14234**	11098	3159	**14257**	**<1%**
$S_{1,1}$	13358	0	**13358**	9832	3530	**13362**	**<1%**
$S_{1,2}$	13354	0	**13354**	9826	3535	**13361**	**<1%**
$S_{1,3}$	13352	0	**13352**	9843	3514	**13357**	**<1%**

TABLE V: `Scenario 2` NPCA-related counters for transmitted and received RTS/MPDUs.

	2. NPCA					
	No of Non-NPCA Tx RTSs	No of NPCA Tx RTSs	No of Non-NPCA Rx RTSs	No of NPCA Rx RTSs	No of Non-NPCA Rx MPDUs	No of NPCA Rx MPDUs
A_1	1368	748	1696	0	90317	0
$S_{1,1}$	1545	0	3985	0	47037	0
$S_{1,2}$	1544	0	4014	0	41247	0
$S_{1,3}$	1556	0	3999	0	48315	0